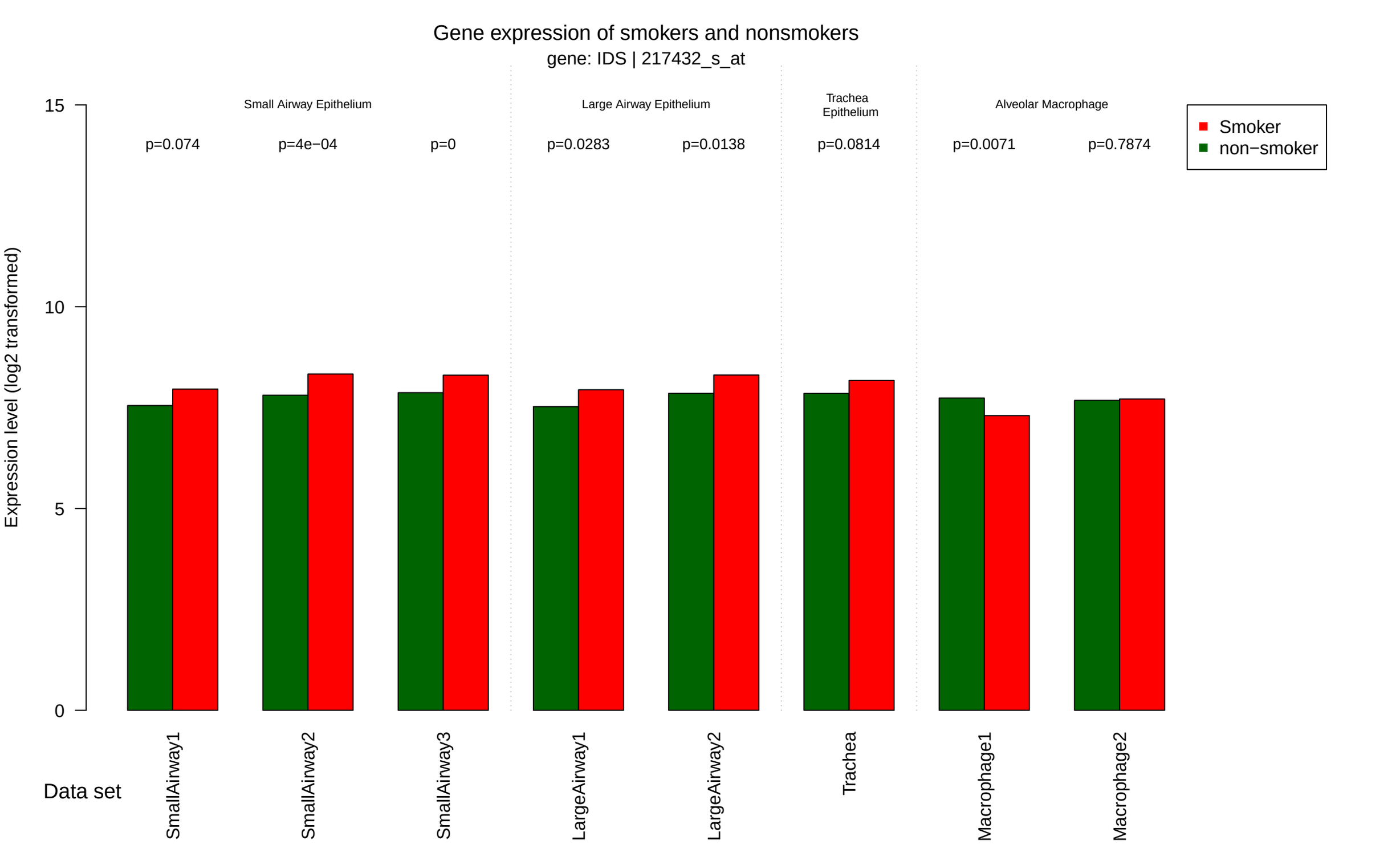
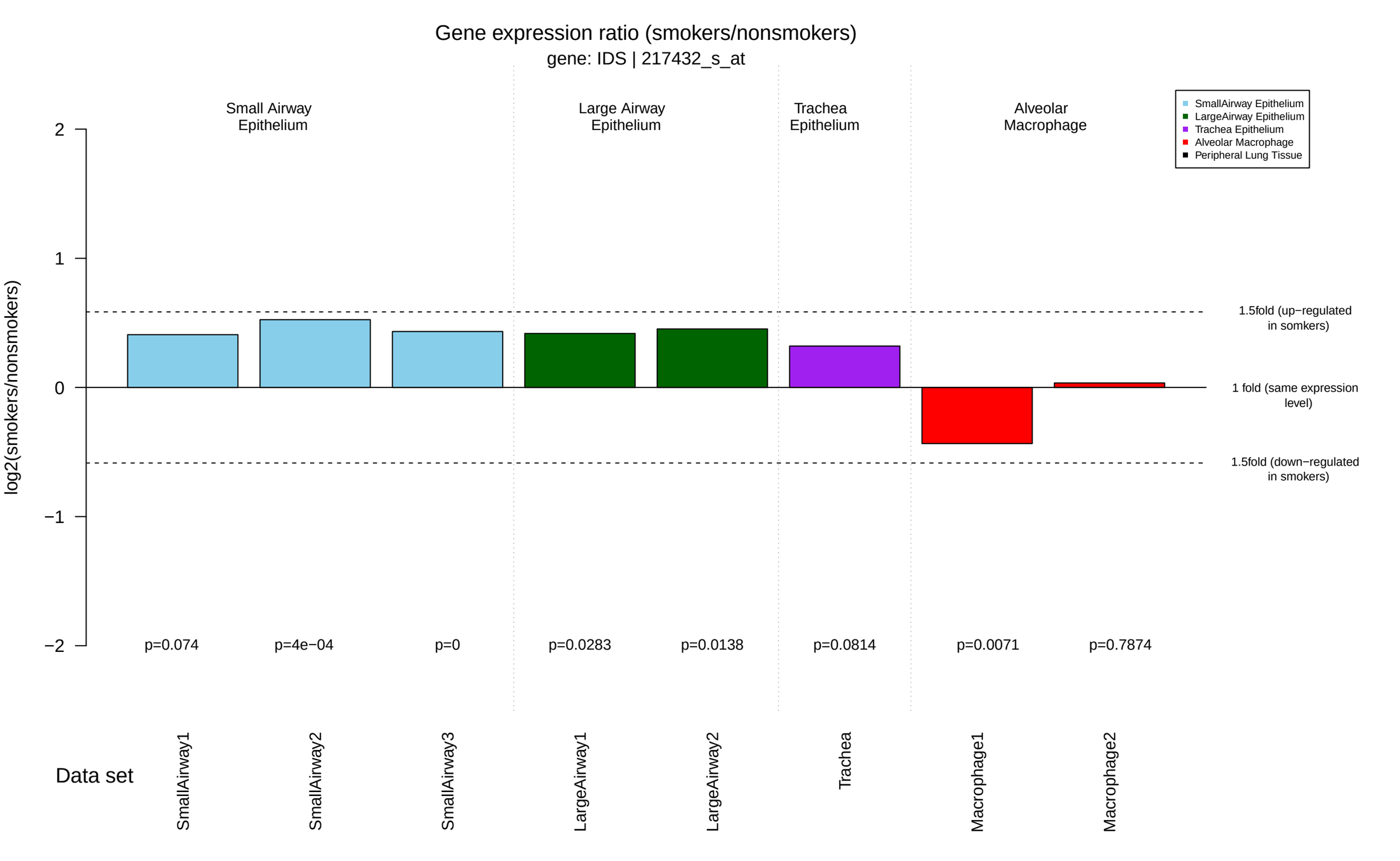
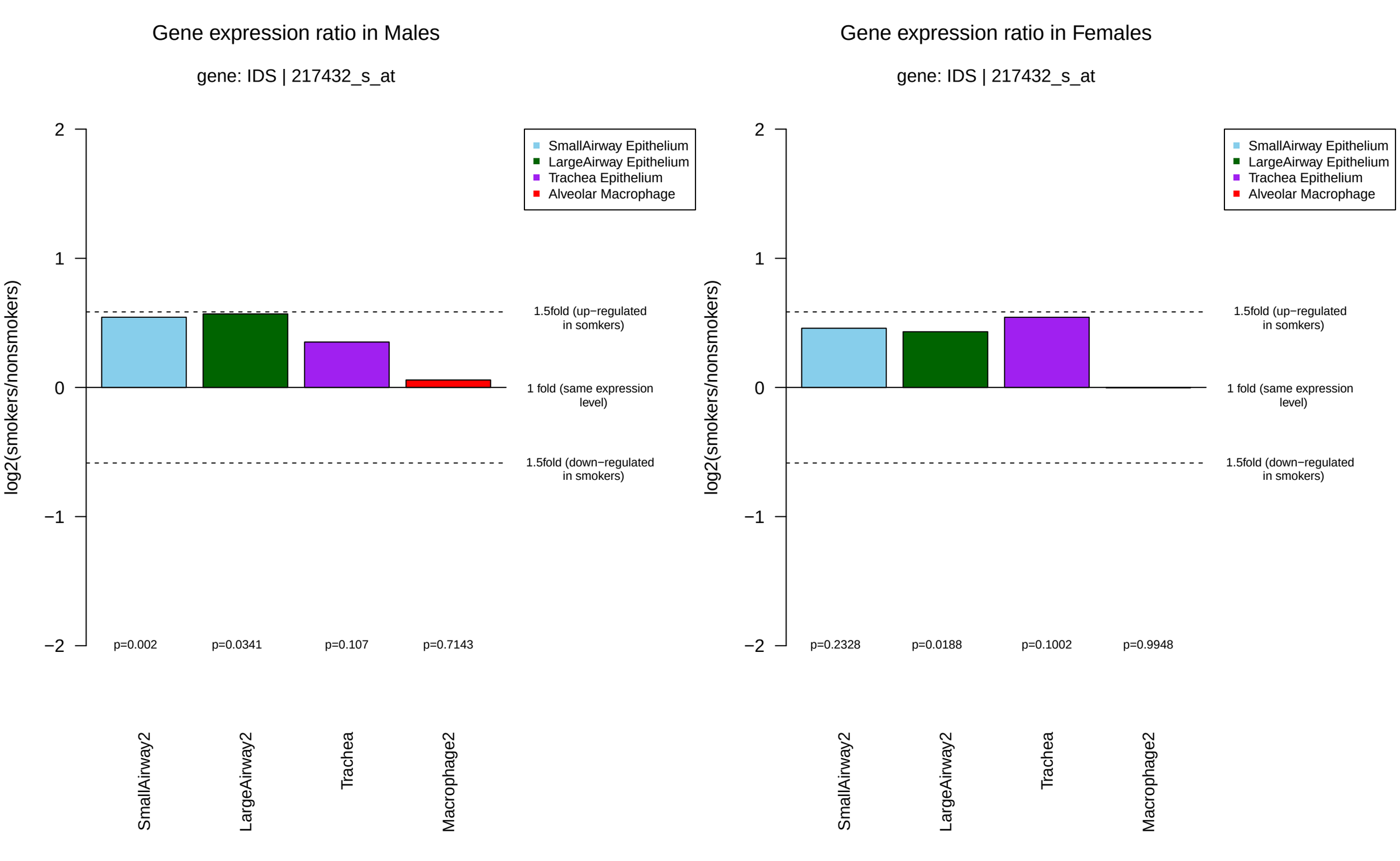
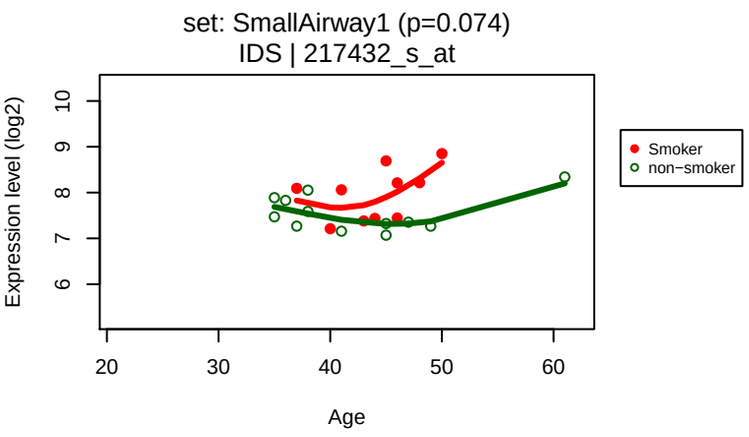
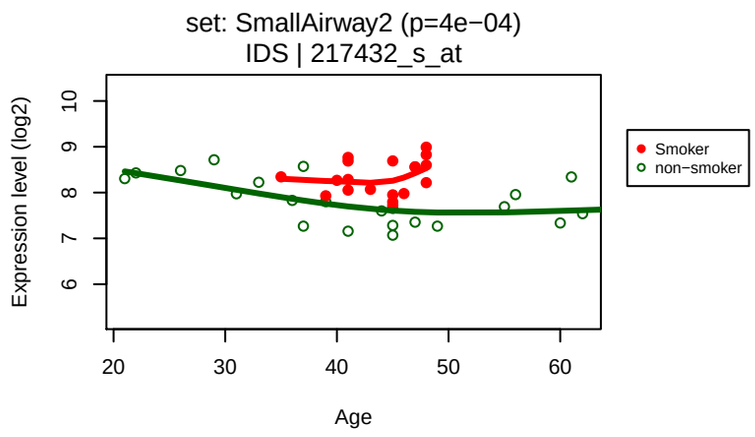
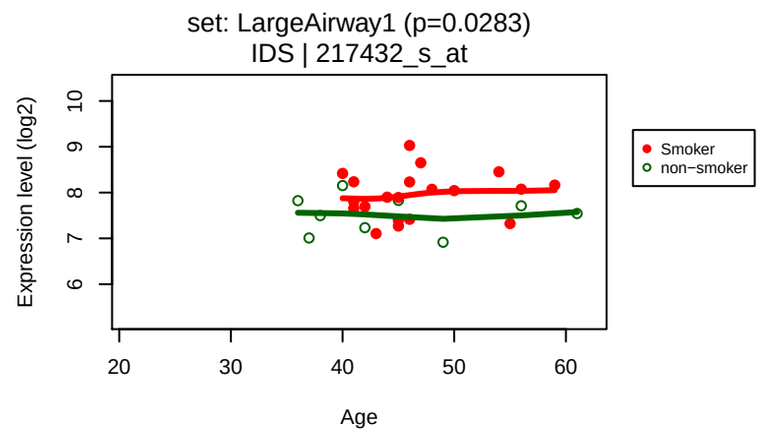

Gene expression of smokers and nonsmokers
gene: IDS | 217432_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.074
p=4e−04
p=0
p=0.0283
p=0.0138
p=0.0814
p=0.0071
p=0.7874
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

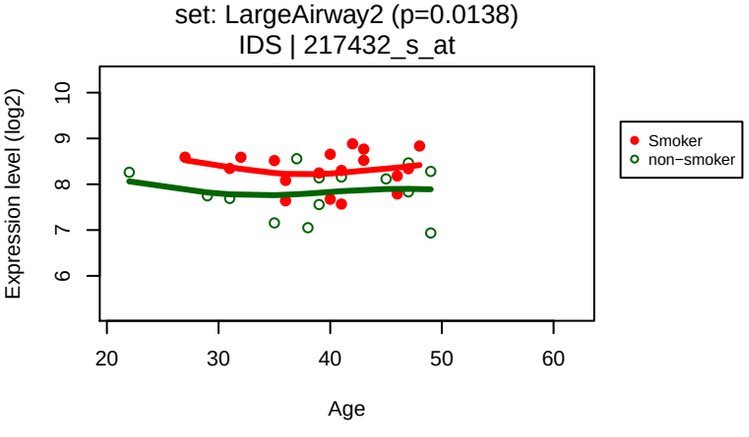

Gene expression ratio (smokers/nonsmokers)
gene: IDS | 217432_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.074
p=4e−04
p=0
p=0.0283
p=0.0138
p=0.0814
p=0.0071
p=0.7874
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

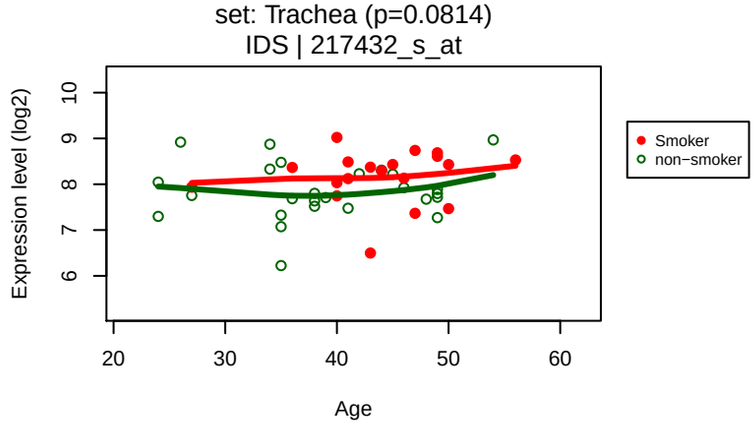

Gene expression ratio in Males
gene: IDS | 217432_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.0341
p=0.107
p=0.7143
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IDS | 217432_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2328
p=0.0188
p=0.1002
p=0.9948
SmallAirway2
LargeAirway2
Trachea
Macrophage2

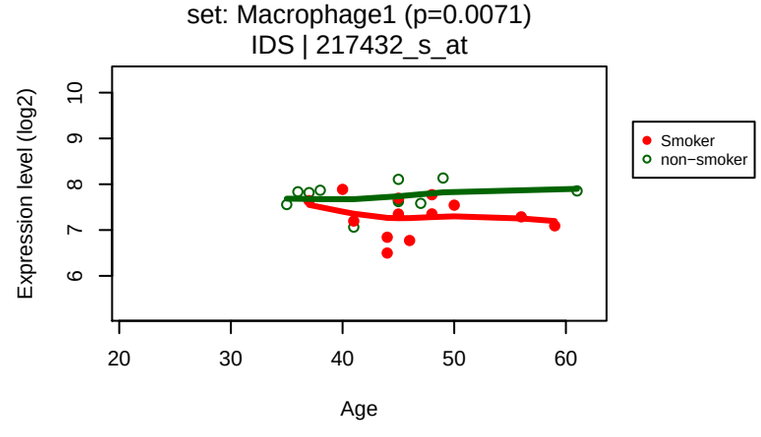

set: SmallAirway1 (p=0.074)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker

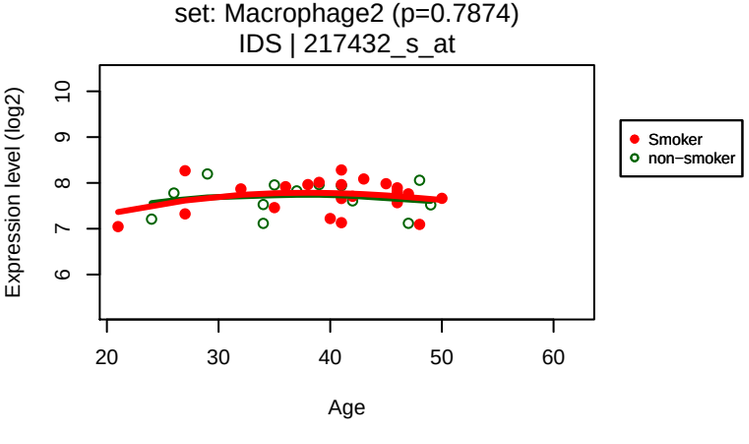

set: SmallAirway2 (p=4e−04)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker

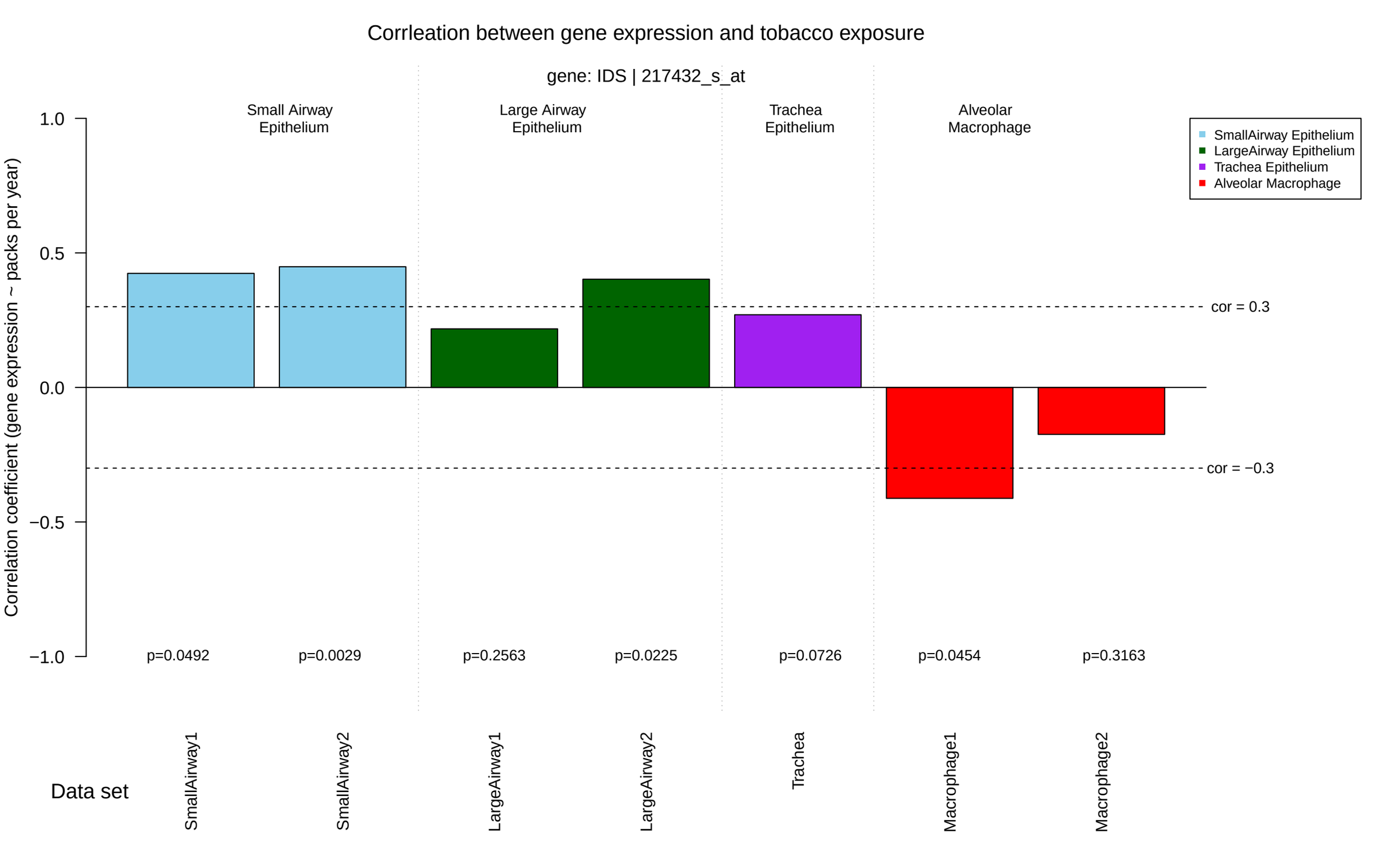

set: LargeAirway1 (p=0.0283)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0138)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0814)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0071)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7874)
IDS | 217432_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IDS | 217432_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0492
p=0.0029
p=0.2563
p=0.0225
p=0.0726
p=0.0454
p=0.3163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2